

AKSHAR VANDARA

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EDUCATION

Northeastern University

Master of Science, Game Science and Design (GPA: 3.87 / 4.0)

May '26

Boston, MA

Gujarat Technological University

Bachelor of Engineering, Computer Engineering (GPA: 3.6 / 4.0)

May '23

Ahmedabad, India

EXPERIENCE

Research Assistant

Feb '25 – Present

Game Design Lab, Northeastern University

Boston, MA

- Engineered an extension for the 'Sturgeon' constraint-solving system, enabling it to simultaneously generate 2D puzzle levels and their corresponding valid solutions.
- Refactored the generation algorithm to achieve a performance increase of over 100x while measurably improving stability compared to prior academic methods.
- Co-authored a research paper detailing the system's architecture and performance metrics, published at the AAAI 2025 AIIDE conference. <https://doi.org/10.1609/aiide.v21i1.36818>
- Currently developing a method to generate 3D game levels by extracting and blending successful procedural patterns from 2D games to the constraint-solving system.

Gameplay Systems & UI Programmer

May '25 – Aug '25

Otisco Studios

Orlando, FL

- Designed and implemented 8+ fundamental gameplay systems, including save/load management, UI, and environmental mechanics, across two distinct titles.
- Developed enemy and NPC AI for both roguelike and open-world genres, creating 5+ distinct combat archetypes using Unreal Engine's Behavior Trees and Blackboards.
- Identified and resolved over 30 critical bugs during the pre-launch phase, ensuring overall product stability for release.
- Conducted extensive code profiling and refactoring across multiple projects, improving the average gameplay framerate by ~100%.

Unity Developer

Mar '23 – Jul '24

Arcadon Games

Bangalore, India

- Engineered core mechanics for Cricket Tycoon, using state machines and player interaction systems, achieving a 30% improvement in gameplay responsiveness.
- Integrated Firebase backend solutions, managing over 1,000+ user data entries, inventory systems, and game progress synchronization with real-time updates via cloud functions.
- Designed AI simulation systems capable of simulating up to 30 randomized matches per minute, enhancing gameplay variety and reducing load times by 40%.
- Conducted alpha testing with 100+ users, achieving 85% positive feedback, and resolved 20+ critical issues to significantly improve game stability and user engagement.

PROJECTS

Spacetime Level Generation [[GitHub](#)] [[Publication](#)] | *Python, SAT Solvers, Pandas*

- Leveraged Python to optimize the 'Sturgeon' constraint-solver, integrating solution-path validation to automate the creation of playable 2D puzzles.
- Increased level generation speed by 100-500 times and improved error-proofing compared to previous methods like STWFC (Space-Time WaveFunctionCollapse)
- Co-authored and published a research paper on the system that was accepted into the 2025 AAAI conference (AIIDE 25).

Red Circuit Rebellion [[Steam](#)] | *Unreal Engine 5, C++, Blueprints, Behavior Trees*

- Developed 5+ core gameplay systems, including save/load, environmental fracturing, health management, and objectives manager.
- Implemented enemy AI using Behavior Trees and EQS to create 3 combat archetypes.
- Scripted 2 game modes (Story and Arcade) with specific objective logic and enemy spawners for each.
- Optimized game assets and blueprints, increasing average FPS by ~100% during gameplay.

Cricket Manager: League [[Playstore](#)] | *Unity, C#, Firebase, Firebase Realtime DB, Firebase Cloud Functions*

- Architected sports management system capable of handling 1000+ concurrent users with Firebase, maintaining sub-100ms response times for pseudo-multiplayer features.
- Orchestrated alpha testing phase with 100+ users, iterating through 3 major versions to achieve 85% user satisfaction and 40% reduction in bug reports.

SKILLS

Game Dev:	Unity, Unreal Engine, Blueprints, Game Physics, Core-gameplay management, Database handling
Game Design:	Animations, Animation Rigging, UI integration, Environment Design, 3D Modeling
Languages:	C#, C++, JavaScript, Python, Java, R, C, SQL
Python Libs:	NumPy, Pandas, PyTorch, Cuda, TensorFlow, PyGames, Flask, Django, SQLAlchemy
Software Dev:	HTML, CSS, JavaScript, Three.js, Tailwind, React, Node, Vue, Angular, Next.js, Vite.js
Dev Tools:	Git, Perforce, Firebase Realtime DB, Firebase Cloud functions, SourceTree, Gitlab, Figma, Trello